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Roll No

EX-302 (GS)

B.E. III Semester Examination, June 2020

Grading System (GS)

Electro-Magnetic Theory

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Define electrostatic potential and electric field intensity. Hence show that, $E = -\nabla V$.
2. A point charge of 6nc is located at the origin in free space. Find V_p if point P is located at $(0.2, -0.4, 0.4)$ and
 - i) $V = 0$ at infinity
 - ii) $V = 0$ at $(1, 0, 0)$
 - iii) $V = 20$ volts at $(-0.5, 1, -1)$
3. Derive Maxwell's equation in
 - i) Differential form
 - ii) Integral form
 - iii) Free space
 - iv) Harmonically time varying fields
4. Derive the conditions at a boundary between two dielectrics having dielectric constants of ϵ_1 and ϵ_2 .
5. Using Biot-Savart law, derive an expression for inductance per unit length of a long coaxial cable with radii of inner and outer conductors as ' a ' and ' b ' ($b > a$).

OR

Describe normal, oblique incidence and skin effect. Given a non-magnetic material having $\epsilon_r = 2.25$ and $\sigma = 10^{-4} \text{ m / mhos}$. Find numerical values at 2.5MHz for loss tangent.
6. In a region of space $D = 500e^{-0.1x^2} a_x \mu\text{C/m}^2$, find the flux ψ crossing the surface, $x = 5\text{m}$, $0 \leq y \leq 3\text{m}$ and $0 \leq z \leq 2\text{m}$.

7. Show that the energy stored in electric field is given by $W_E = \frac{1}{2} \int_{vol} \rho_v V dv$.

OR

Explain method of images with its applications.

8. State and explain the followings:
 - i) Divergence theorem
 - ii) Stoke's theorem

OR

Express the field $E = 2xyz I_x - 5(x+y+z) I_z$ in cylindrical coordinates and calculate $|E|$ at point P ($r = 2$, $\phi = 60^\circ$, $z = 3$)